Parameterizing the Genetic Architecture Under Stabilizing Selection

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Motivation

1. Figure adapted from Yengo et al. (2022).

Main idea

- We derive β_j's from evolutionary theory.
- Start from stabilizing selection under Fisher's geometric model (FGM) (Fisher 1930; Tenaillon 2014):

$$\text{fitness} \propto \exp\left(-\frac{\|z\|^2}{2V_S}\right)$$

- Introduce a focal trait y that is coupled to an unobserved selected trait z.
- Derive the induced frequency dependence of β_j.

Evolutionary setup

Stabilizing selection

- Fitness is modeled as

$$\text{fitness} \propto \exp\left(-\frac{\|z\|^2}{2V_S}\right)$$

- For locus j with effect vector α_j, the marginal selection coefficient is

$$s_{\text{eff}}(x_j) = \left(x_j - \frac{1}{2}\right)\frac{\|\alpha_j\|^2}{V_S}$$

- This yields a stationary allele-frequency density that depends on $\|\alpha_j\|^2/W_S$.

From fitness effects to focal-trait effects

- We model the focal-trait variance conditional on the fitness effect by

$$\mathbb{E}[\beta_j^2 \mid \|\alpha_j\|] = \sigma_b^2(1-\rho_{ab}^2) + \rho_{ab}^2 \frac{\sigma_b^2}{\sigma_a^2} \|\alpha_j\|^2$$

- Here:
  - σ_a^2 controls mutational variance for the selected trait,
  - σ_b^2 controls the overall variance scale of the focal trait,
  - ρ_{ab}^2 controls coupling between the focal trait and selection.

Approximate genetic architecture

Key derivation

- Using Bayes' rule together with the stationary allele-frequency model,

$$p(\|\alpha_j\|^2 \mid G_j) \propto p(G_j \mid \|\alpha_j\|^2) p(\|\alpha_j\|^2)$$

- Under the low-mutation approximation,

$$\mathbb{E}[\|\alpha_j\|^2 \mid G_j] = \frac{\sigma_a^2}{1 + 2\frac{\sigma_a^2}{W_S}\hat{p}_j(1-\hat{p}_j)}$$

- Substituting this into the pleiotropy model gives

$$\mathbb{E}[\beta_j^2 \mid G_j] = \sigma_b^2 \left(1 - \rho_{ab}^2 \frac{2\frac{\sigma_a^2}{W_S}\hat{p}_j(1-\hat{p}_j)}{1 + 2\frac{\sigma_a^2}{W_S}\hat{p}_j(1-\hat{p}_j)}\right)$$

Interpretation

- The model is frequency-dependent, but unlike the α-model it does not diverge as $\hat{p}_j \to 0$.
- The parameters have direct evolutionary interpretations.
- In practice, the most identifiable selection-sensitive quantity is the product

$$\rho_{ab}^2 \sigma_a^2 / W_S$$

Simulation design

How these models are usually evaluated

- Most benchmarking studies condition on a fixed genotype matrix rather than simulating evolution explicitly.

$$y = G\beta + \varepsilon$$

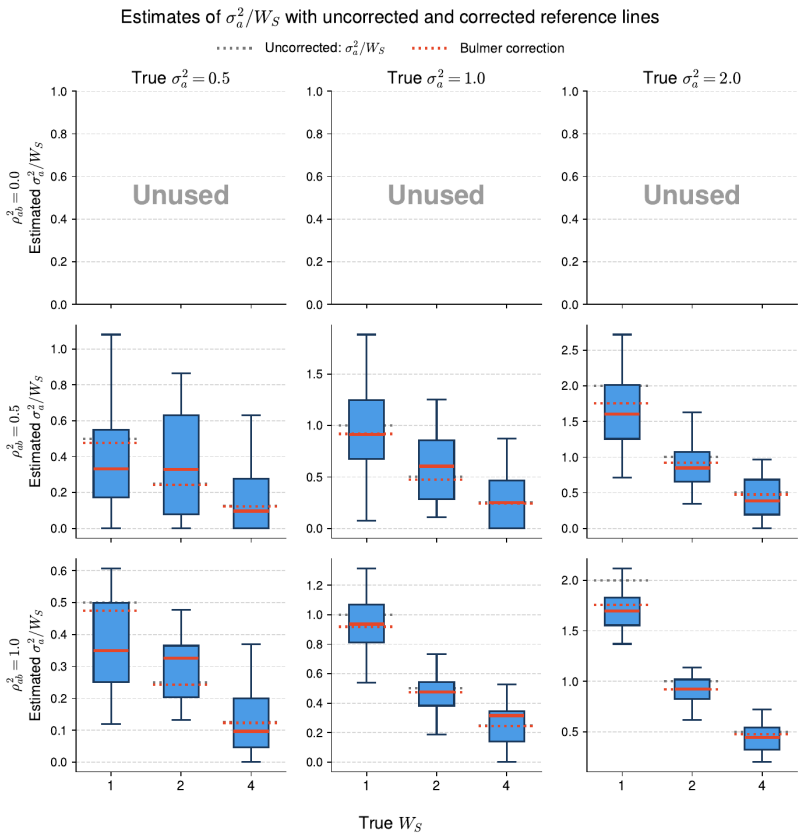
- They then generate phenotypes by varying only:
  - the random effect sizes under a chosen prior of β
  - the residual noise ε
- You simulate from your model so can never be wrong.

Forward simulations in SLiM

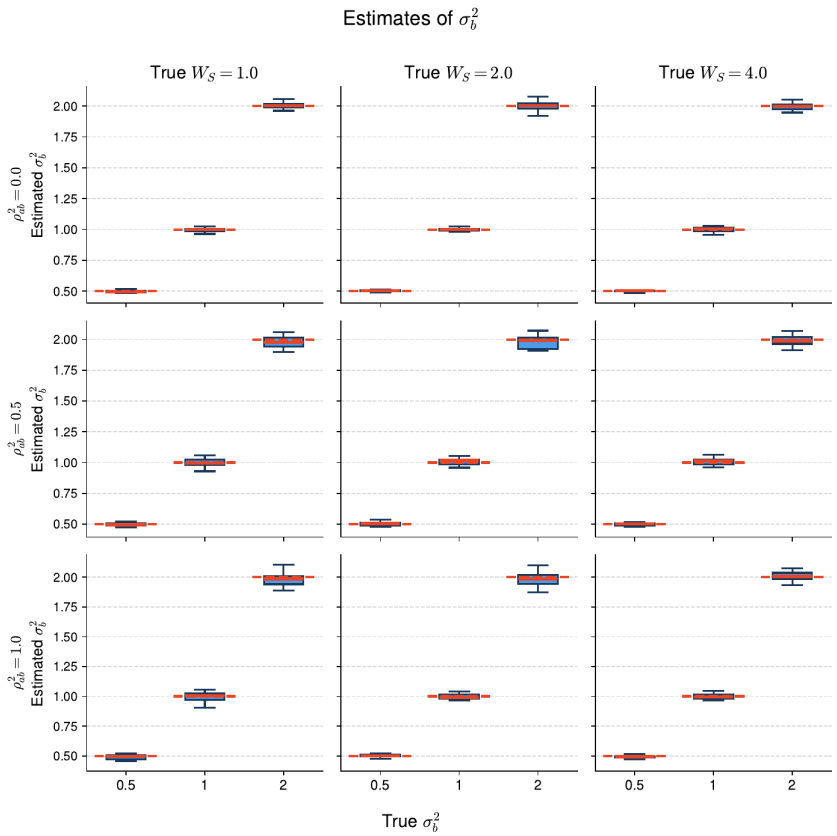
- Individual-based forward simulations under stabilizing selection.
- Constant diploid population size $N = 10{,}000$.
- Genome length $L = 2 \times 10^6$, mutation rate $\mu = 10^{-8}$, recombination rate $r = 10^{-5}$.
- Grid over:
  - $\sigma_a^2 \in \{0.5, 1.0, 2.0\}$,
  - $W_S \in \{1.0, 2.0, 4.0\}$,
  - $\rho_{ab}^2 \in \{0, 0.5, 1\}$.
- We evaluate both variance-component recovery and held-out BLUP prediction.

Results

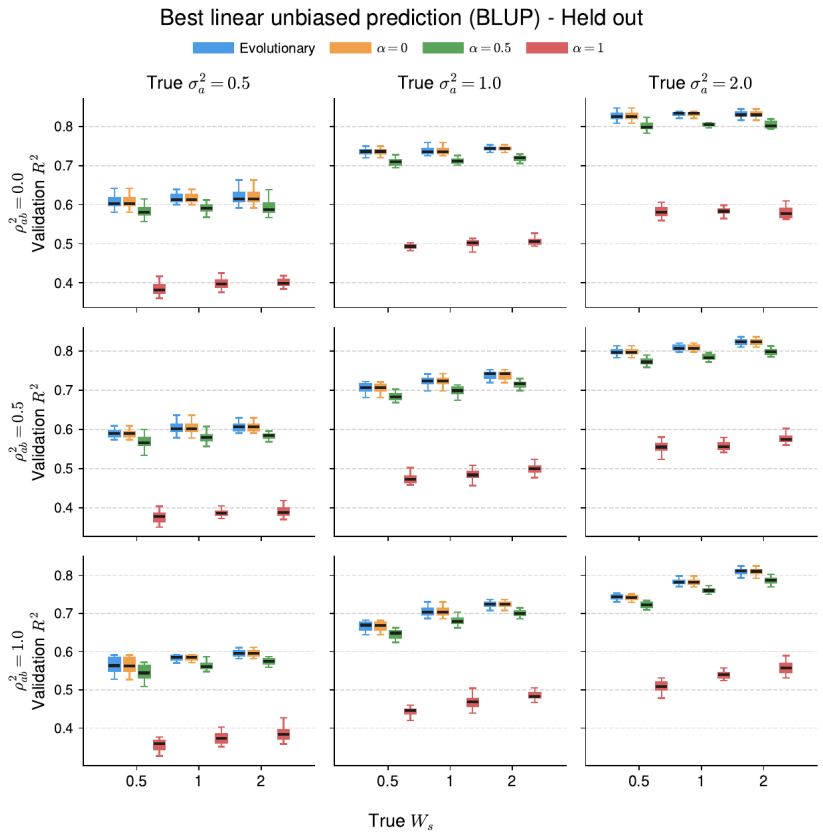
Recovering the selection-sensitive component



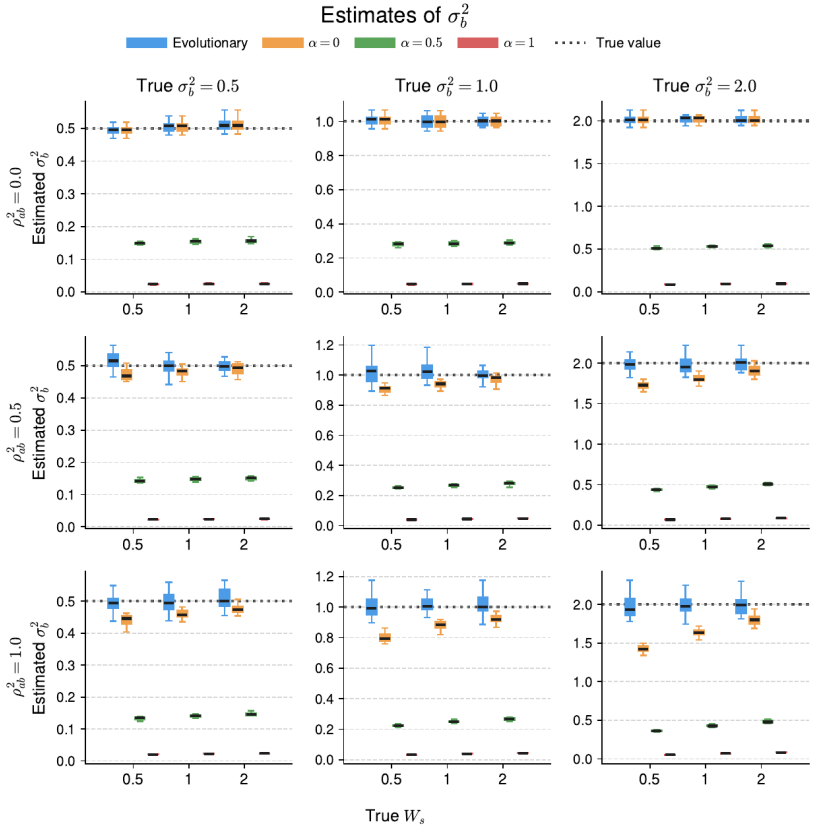
Recovering the overall variance scale



Prediction accuracy



Variance components



Summary

- We derive a frequency-dependent mixed model for effect sizes from stabilizing selection rather than imposing it heuristically.
- The framework connects fitness-landscape theory to REML estimation and BLUP prediction.
- We can do statistical genetics like physics instead of random curve fitting.
- In simulations, the model recovers the focal-trait variance well and generally outperforms conventional α-model baselines in prediction.
- Extending the framework to richer pleiotropy models and explicit linkage disequilibrium is a natural next step.

References

Fisher, Ronald Aylmer. 1930. *The Genetical Theory of Natural Selection*. Oxford University Press. https://openlibrary.org/books/OL6898400M/The_genetical_theory_of_natural_selection.

Tenaillon, O. 2014. "The Utility of Fisher's Geometric Model in Evolutionary Genetics." *Annual Review of Ecology, Evolution, and Systematics* 45 (1): 179–201. https://doi.org/10.1146/annurev-ecolsys-120213-091846.

Yengo, Loïc, Sailaja Vedantam, Eirini Marouli, Julia Sidorenko, Eric Bartell, Saori Sakaue, Marielisa Graff, et al. 2022. "A Saturated Map of Common Genetic Variants Associated with Human Height." *Nature* 610 (7933): 704–12. https://doi.org/10.1038/s41586-022-05275-y.